



Novel thermoelectric materials: free-standing poly(ϵ -caprolactone)/poly(*m*-anthranilic acid)/Fe₂O₃, Fe₃O₄ nanofibers

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Abstract

Owing to global warming and the shortage of fossil fuel reserves, there has been a search for sustainable and cleaner energy sources. Thermoelectric (TE) generators, which are environmentally friendly alternatives using the Seebeck effect, can generate electricity with some of the waste heat and contribute to slowing the rate of global warming. It can be said that thermoelectric polymer materials have become more attractive than inorganic thermoelectric materials due to their high mechanical elasticity, non-toxicity, low cost, easy processability, and poor thermal conductivity. However, since the thermoelectric properties alone are not sufficient, the thermoelectric properties can be increased by adding conductive polymers, organic, inorganic, or polymeric nanoparticles. In this study, free-standing and flexible nanofiber composites were obtained with poly(ϵ -caprolactone) (PCL), poly(*m*-anthranilic acid) (PANA), Fe₂O₃, and Fe₃O₄ by the electrospinning method. The TE properties of the obtained nanofiber were investigated. The highest Seebeck coefficient and thermoelectric power factor were obtained from PCL/PANA/Fe₃O₄ nanofiber composite as 49 μ V/K and 0.13 μ W/mK², respectively. Nanofibers were characterized by XRD and FTIR, and their surface morphology was examined by SEM.

Keywords Nanofiber · Flexible thermoelectric materials · Poly(*m*-anthranilic acid)

Introduction

The fact that most of the energy given to the energy conversion systems used today is given to the atmosphere as waste heat causes global warming (Li et al. 2019; Forman et al. 2016; Kim et al. 2018; Cullen and Allwood 2010). Due to global warming and the scarcity of fossil fuel reserves, cleaner and more sustainable energy sources are sought. Thermoelectric (TE) generators, which are environmentally friendly alternatives, can help to slow the rate of global warming by converting some of this waste heat into electricity (Du et al. 2012; Juntunen et al. 2018; Kroon et al. 2016; Mori and Priya 2018). These can generate electricity using the Seebeck effect from heat emitted from sources such as motor vehicles, computers and even the human body, as well as retainable heat sources such as solar energy and natural

hot springs. In the last two decades, with the emergence of new ideas for the use of small-scale TE devices in daily life, intensive research has been carried out for new materials with improved TE properties. It can be said that thermoelectric polymer materials have become more attractive than inorganic thermoelectric materials due to their high mechanical elasticity, non-toxicity, low cost, easy processability, and poor thermal conductivity (Wang et al. 2014, 2011; Huner and Karaman 2022, 2018; Yan et al. 2001; Yan and Toshima 1999; Xuan et al. 2010; Dubey and Leclerc 2011; Bubnova et al. 2011; Zhang et al. 2014). However, since the thermoelectric properties alone are not sufficient, the thermoelectric properties can be increased by adding conductive polymers, organic, inorganic, or polymeric nanoparticles (Wang et al. 2017; Chatterjee et al. 2013; Huner 2022).

Conventional thermoelectric generators (TEGs) are mostly prepared in the form of cubic-shaped blocks. The disadvantage of this form is that they do not have good contact with complex heat sources (Kim et al. 2018). Flexible TE materials do not have this problem, and TE properties can be quite good (Fang et al. 2017). Although there were few studies (Li et al. 2004; Chronakis et al. 2006) in the past when the literature was searched, studies on the TE properties of

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flexible thin films have increased in recent years (Song et al. 2021; Cataldi et al. 2020; Okada et al. 2020; Bharti et al. 2021; Hu et al. 2021; Dong et al. 2021). Electrospinning to obtain flexible thin films has some advantages, such as being simple and easy to control. Electrospinning is a good method to create randomly distributed pores in nanofibers. It is known that the thermal conductivity of porous structures decreases; they can adapt to deformation, thus making a significant contribution to flexibility and ductility (Jin et al. 2018; Yang et al. 2016).

Poly(ϵ -caprolactone) (PCL), which has very good mechanical properties, is a biodegradable and biocompatible polymer and is used in packaging materials, tissue engineering, and biomaterials (Sheikh et al. 2014; Gungor-Ozkerim et al. 2014; Huner et al. 2023; Luong-Van et al. 2007). In addition, its nanofibers obtained by electrospinning have been used in many studies (Polat et al. 2016; Huner 2023). Poly(m-anthranilic acid) (PANA) is a conjugated polymer used in tissue engineering applications with its ability to reduce hydrophobicity and obtain a carboxylated surface (Congiu et al. 2005). Iron oxide nanomaterials are used in biomaterials, such as the use of Fe_2O_3 , Fe_3O_4 , in magnetic bioactive glasses to release drugs for cancer cells (Singh and Srinivasan 2011).

Although rare earth elements such as Bi_2Te_3 and PbTe are still among the materials with the best thermoelectric properties, the aim of this study is to present non-toxic materials and alternative thermoelectric materials with flexible good thermoelectric properties by using a cost-effective synthesis method. PCL/PANA and PCL/PANA/ Fe_3O_4 nanofiber composites have not been reported for thermoelectric materials, and in this study, nanofiber composites were prepared using simple processes and their structural, morphological, and thermoelectric properties were investigated.

Experimental

Materials

Poly(ϵ -caprolactone) (PCL; Mw: 80.000; Polyscience), m-aminobenzoic acid (Sigma Aldrich), ammonium persulfate (APS; Merck), sodium hydroxide (NaOH; Merck), hydrochloric acid (HCl; Merck), dimethylformamide (DMF; Merck), chloroform (CHCl_3 ; Merck), iron(III) oxide, and iron oxide (II,III) (Fe_2O_3 , Fe_3O_4 ; Sigma Aldrich) were used experimentally.

Synthesis of PANA

PANA synthesis is the same as in the literature (Polat et al. 2016). Accordingly, a solution of 0.04 mol of ammonium persulfate in water was added dropwise to a solution of

3-aminobenzoic acid (0.04 mol) dissolved in NaOH (0.5 M), and then it was stirred for 24 h at room temperature. The filtered precipitate was washed with HCl and water and at 60 °C dried.

Fabrication of PCL/PANA/ Fe_2O_3 , Fe_3O_4 nanofibers PCL (20% w/v) was dissolved in the DMF and CHCl_3 mixture (1/1 v/v; 5 mL), and PANA was added as 4 and 10% (w/w) of the PCL and Fe_2O_3 and Fe_3O_4 as 5% (w/w) of the PCL. These mixtures were stirred to obtain a homogeneous solution for 3 h at room temperature. Electrospinning solutions placed into a 0.7 mm outer diameter syringe needle connected to a high voltage direct current power source were electrospun horizontally with a collection distance of 15 cm on aluminum foil at a drive voltage of 15 kV. The flow rate of the mixture controlled was 1 mL/hr by a syringe pump. Power supply was used ES 30 Model; Ormond Beach, Gamma High Voltage Inc. for electrospinning.

Characterization

The structure of nanofibers was characterized by X-ray diffraction (XRD, Rigaku D/MAX-Ultima + /PC) and attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR, TENSOR27, Bruker Optic GmbH). Electrical conductivity measured using four probes and Seebeck coefficient were measured technique at room temperature with ENTEK FPP 470 and ENTEK (SC 001), respectively. Thermoelectric power factor ($\text{PF} = \sigma S^2$, S: Seebeck coefficient, σ : electrical conductivity) was calculated by using electrical conductivity and Seebeck coefficient (Zhu et al. 2017). The morphology of nanofibers was investigated by SEM (Zeiss EVO® LS 10). The average fiber diameters of fibers were measured from the SEM images using ImageJ software.

Results and discussion

Various mixtures were prepared with PANA, Fe_2O_3 , and Fe_3O_4 by keeping the PCL ratio constant in DMF and CHCl_3 (1:1), and nanofiber films were obtained from the prepared mixtures by the electrospinning method. The codes given to nanofiber films and their contents (percentage by weight) are given in Table 1.

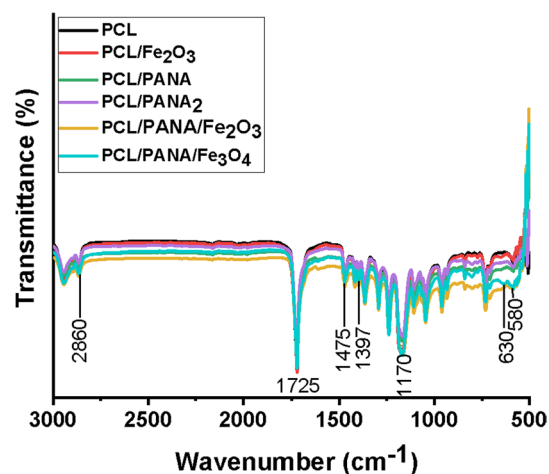
Figure 1 shows digital photographs of free-standing nanofiber film obtained from the PCL/PANA/ Fe_3O_4 sample. Other nanofiber films not shown in the photo obtained by electrospinning also show excellent flexibility and can be easily bent without cracking. Conductive polymers hold great promise for thermoelectric generators, but are too hard and brittle (Huner 2024) to be seamlessly integrated into wearable devices for effective energy harvesting. Therefore, it is important to obtain structures that are both conductive and soft (Kim et al. 2020). Adding non-volatile plasticizers

Table 1 The codes given to the samples and their contents (percentage by weight)

Sample	PCL (w%)	PANA (w%)	Fe ₃ O ₄ (w%)	Fe ₂ O ₃ (w%)
PCL	20	–	–	–
PCL/Fe ₂ O ₃	20	–	–	5
PCL/PANA	20	10	–	–
PCL/PANA ₂	20	4	–	–
PCL/PANA/Fe ₂ O ₃	20	10	–	5
PCL/PANA/Fe ₃ O ₄	20	10	5	–

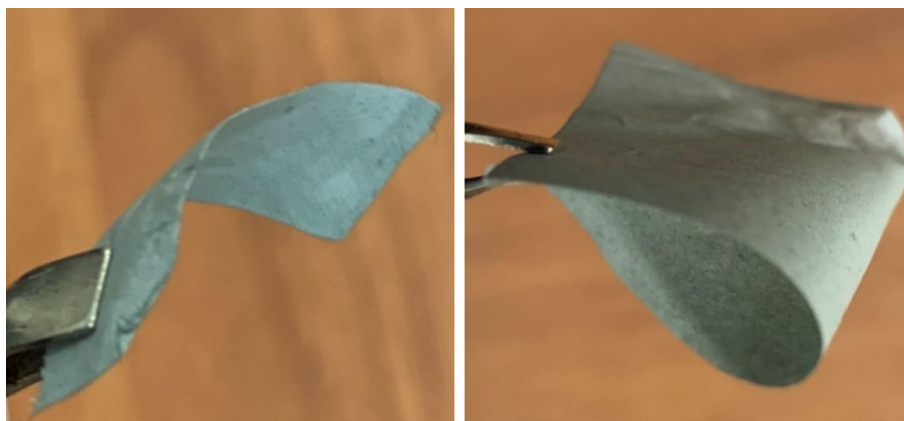
to conductive polymers, such as surfactants containing polyethylene glycol segments, polyols, and ionic liquids, is also used for this purpose. In some studies, soft films have been obtained with Poly(3,4-ethylenedioxythiophene) (PEDOT) which has optical transparency, high stability, medium band gap, and low redox potential in the conducting state (Huner and Sarac 2021), and their potential for use in wearable devices has been reported. An ultrathin PEDOT was produced by a micelle soft patterning method and the thermoelectric performance of the produced film was investigated after some treatments. These PEDOT films, which have a 54% improvement in performance compared to traditional preparation methods, have the potential to be used in wearable devices (Cao et al. 2021). Here, flexible films were obtained by using the electrospinning method, which is a convenient and simple method to prepare.

ATR-FTIR spectra of nanofiber films are given in Fig. 2. The characteristic peaks of PCL are peaks at about 2860, 1725, 1397, and 1170 cm⁻¹, which are attributed to C–H stretching, C=O stretching, C–H bending, and C–O stretching, respectively. The peaks attributed to C=C stretching and –N–H stretching at 1567 and 1507 cm⁻¹, respectively, belong to PANA. The peak of Fe₃O₄ and Fe₂O₃ (Yinsheng et al. 1998) nanoparticles was observed at around 580–630 cm⁻¹. In another study with PANA in the literature

**Fig. 2** ATR-FTIR spectrum of nanofibers

(Giray et al. 2013), weak peaks were seen for –N–H and C=C peaks. Due to the strong peaks of PCL, these peaks were weak and could not be seen clearly in PANA nanofibers. This may be due to the low concentration of PANA and iron oxide nanoparticles in the nanofibers and their inhomogeneity in the measured area.

The XRD patterns of the nanofiber mats are given in Fig. 3. According to the XRD patterns of Fe₂O₃ nanoparticles, it is seen that there are distinct peaks at approximately 2θ ≈ 33° and 36° (Zieliński et al. 2010; Tong et al. 2014). According to the XRD patterns of Fe₃O₄ nanoparticles, it is seen that there are distinct peaks at approximately 2θ ≈ 31.2, 36.5, 44.3, 54.5, 58.1, and 63.9° (Delice et al. 2024). It showed sharp peaks, especially around 21.4° and 23.8°, for the crystal structure of PCL (Castilla-Cortázar et al. 2019). It was observed that the XRD spectra of PCL/PANA, PCL/PANA₂, PCL/PANA/Fe₂O₃, and PCL/PANA/Fe₃O₄ nanofibers had peaks at approximately 21.4, 23.8, 38.6, and 44.5°. Accordingly, it is suggested that the peaks at 21.4° and 23.8° observed in all nanofibers should be attributed to the crystalline structure of PCL, and the peaks at

Fig. 1 Photographs of flexible and free-standing PCL/PANA/Fe₃O₄ nanofiber film

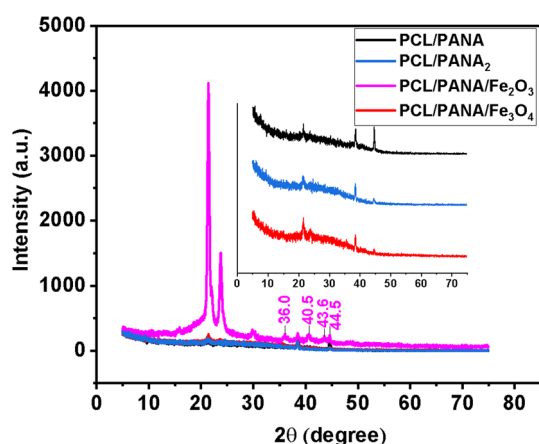


Fig. 3 The XRD patterns of nanofibers

38.6 and 44.5° should be attributed to the crystalline structure of PANA. When the XRD patterns of PCL/PANA and PCL/PANA₂ nanofibers are compared, it is seen that the intensity of the peaks at 38.6° and 44.5° attributed to PANA is higher in the PCL/PANA nanofiber with a higher amount of PANA, as expected. On the other hand, it is seen that the intensity of the peaks indicating the crystalline structure of PCL is higher in the PCL/PANA₂ nanofibers.

Although the intensities are very low for PCL/PANA/Fe₃O₄ and especially PCL/PANA/Fe₂O₃ nanofibers, the small peaks observed at 30.2, 36.0, 40.5, and 43.6° in the PCL/PANA/Fe₂O₃ nanofiber can be attributed to iron oxide nanoparticles. Although the intensity of the peaks attributed to the crystalline structure of PCL is very high in the PCL/PANA/Fe₂O₃ nanofiber, the peaks attributed to both PANA and Fe₂O₃ can be seen. The intensity of the characteristic peaks attributed to the crystalline structure of PCL may have caused the peaks of PANA and iron oxide nanoparticles to appear weakly or to disappear, as in another study (Abdelrazek et al. 2016). As a result, the effect of PANA on the crystalline structure of PCL is seen more clearly, while the effect of Fe₂O₃ and especially Fe₃O₄ nanoparticles could not be seen very clearly for the amount of these nanoparticles used.

The Mott formula (Huner 2024) is often used to design organic compounds with good thermoelectric properties and to interpret the thermoelectric properties of organic thermoelectrics. This formula, given in Eq. (1), relates S to conductivity. S also depends on the slope of the density of states (DoS) at the Fermi level, the steeper the slope, and the higher the Seebeck coefficient.

$$S(E, T) = \frac{\pi^2}{3} \cdot \frac{k^2 T}{q} \cdot \left(\frac{\partial \ln(\sigma(E))}{\partial E} \right)_{E=E_f} \quad (1)$$

Here $\sigma_E(E)$ is a transport function at a given energy level, E , kB is the Boltzmann constant, and T is the temperature.

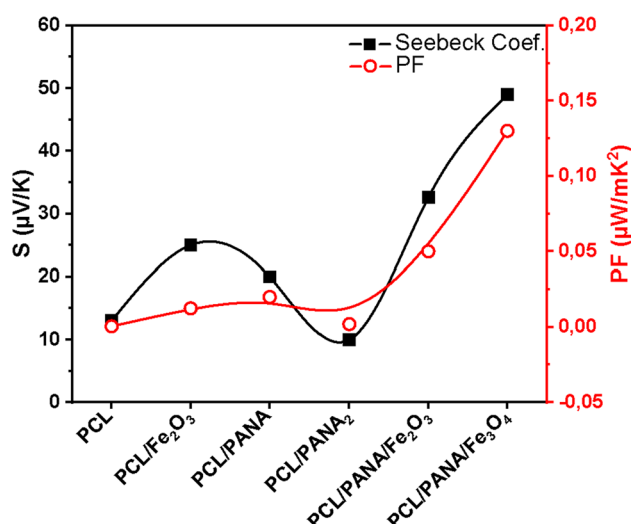


Fig. 4 Seebeck coefficient and the thermoelectric power factor of nanofibers

The transport function E should vary weakly around the chemical potential in the range of a few kBT . The validity of the Mott formula for organic materials, especially conjugated polymers, is debatable due to the non-crystalline nature of conducting polymers, i.e., the structural disorder found in conducting polymers.

Nanofibers whose thermoelectric properties have been studied are not very common in the literature. Some of the studies carried out are as follows:

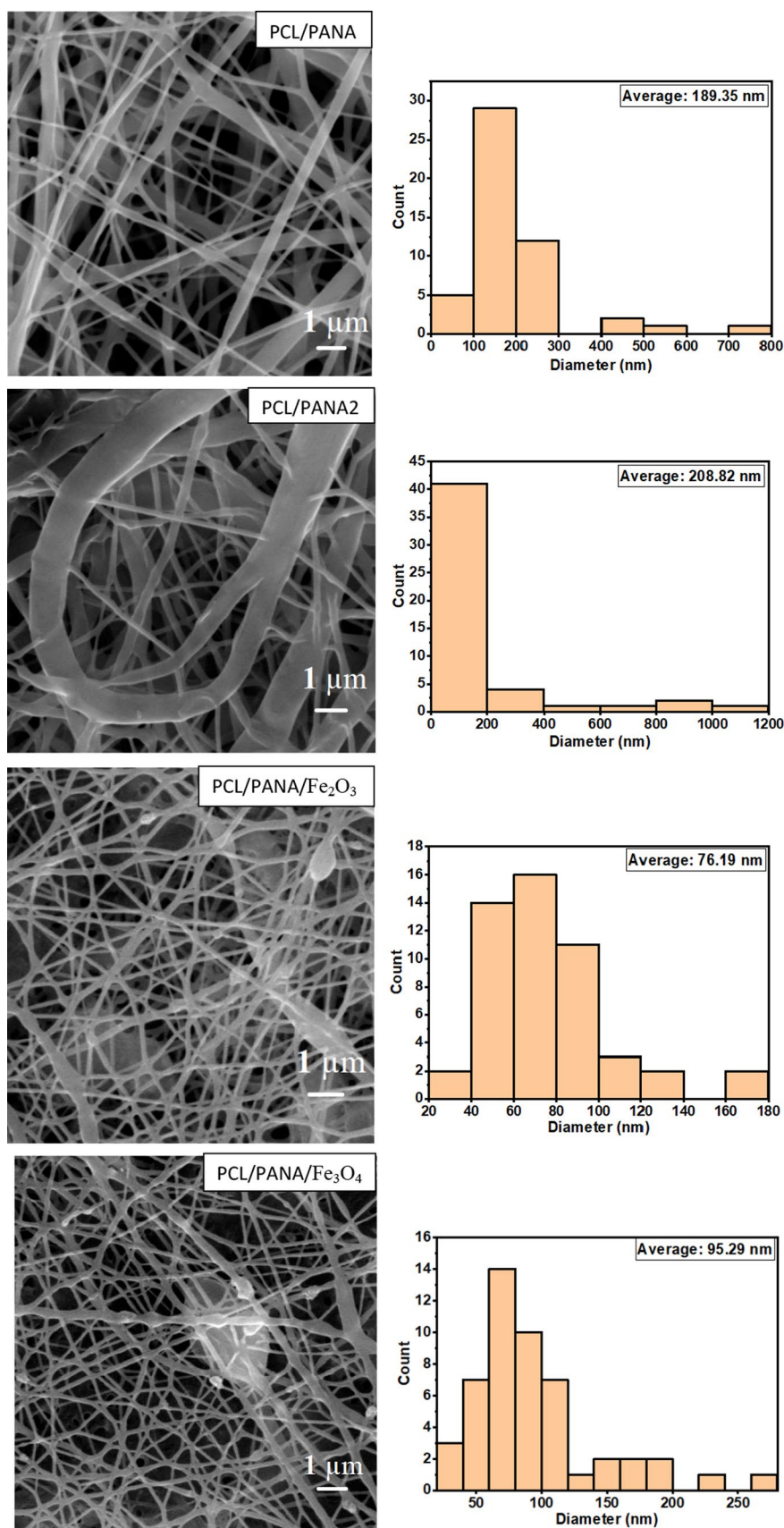
The highest value of S and PF of nanofibers obtained by adjusting the molar ratios of SDS, oxidizing agent $FeCl_3$ and EDOT monomer as given in $n_{SDS} : n_{FeCl_3} : n_{EDOT}$ was reported as 17.4 $\mu V/K$ and 7.2 $\mu W/mK^2$, respectively, by Song et al. (Song et al. 2021). S and PF of composites of carbon nanofibers in ethylene-octene copolymer were reported as 14.2 $\mu V/K$ and $4.2 \times 10^{-5} \mu W/mK^2$, respectively, by Slobodian et al. (Slobodian et al. 2013). Jin et al. have reported the highest S and PF of nanofiber film of PEDOT:PSS/PVA@Ag as 25.1 $\mu V/K$ and 1.2 $\mu W/mK^2$, respectively (Jin et al. 2019).

Figure 4 shows the S and PF of nanofibers. The S increased from 10 to 20 $\mu V/K$ when the PANA ratio

Table 2 Electrical conductivity of nanofibers

Sample	Electrical conductivity (S/cm)
PCL	0.000003 ± 0.000002
PCL/Fe ₂ O ₃	0.20 ± 0.06
PCL/PANA	0.50 ± 0.04
PCL/PANA ₂	0.21 ± 0.13
PCL/PANA/Fe ₂ O ₃	0.45 ± 0.10
PCL/PANA/Fe ₃ O ₄	0.55 ± 0.19

Fig. 5 SEM image (left) and their diameter distribution (right) of nanofibers



increased from 4 to 10%. When 5% Fe_2O_3 was added to the medium with 4% PANA and 20% PCL, the S increased from 10 to 32.6 $\mu\text{V/K}$. The highest S was obtained from PCL/PANA/ Fe_3O_4 nanofiber as 49 $\mu\text{V/K}$. This value is approximately 2.5 times greater than the S of 20 $\mu\text{V/K}$, which is the PCL/PANA nanofiber without Fe_3O_4 , and it is approximately 2–3 times greater than the values of 17.4 (Song et al. 2021), 14.2 (Slobodian et al. 2013), and 25.1 (Jin et al. 2019) in the literature. It can be said that the PF of nanofibers has a similar tendency to the S . The highest PF was obtained from PCL/PANA/ Fe_3O_4 as 0.13 $\mu\text{W/mK}^2$. This value is smaller than the PFs of 7.2 (Song et al. 2021) and 1.2 $\mu\text{W/mK}^2$ (Jin et al. 2019) obtained from nanofibers in the literature but greater than 4.2×10^{-5} $\mu\text{W/mK}^2$ (Slobodian et al. 2013).

Table 2 shows the electrical conductivity of the nanofibers. Electrical conductivity measurements are given as the average of the values measured from 3 different regions of the nanofiber mats with standard deviation values. The electrical conductivity of nanofiber with 4% PANA is higher than that of nanofiber with 5% Fe_2O_3 . The conductivity of PCL/PANA and PCL/PANA/ Fe_2O_3 nanofibers with the same percentage of PCL and PANA decreases from 0.5 to 0.45 S/cm with the effect of Fe_2O_3 , while it increases to 0.55 S/cm with the effect of Fe_3O_4 .

Figure 5 shows an SEM image of nanofibers. In the PCL/PANA sample, it consists of fibers that are less than 100 nm and are microns in size. It was seen that PCL/PANA consisted of fibers with an average of about 189.35 nm in the distribution of nanofiber diameter. In the PCL/PANA₂ sample, which has a lower percentage of PANA than PCL/PANA, it is seen that fibers with higher thickness are more common. Accordingly, PCL/PANA₂ is composed of fibers with a more heterogeneous diameter than PCL/PANA, and it can be said to consist of fibers with an average diameter of about 208.82 nm. Fe_2O_3 and Fe_3O_4 particles in PCL/PANA/ Fe_2O_3 and PCL/PANA/ Fe_3O_4 nanofibers are observed on fibers with average diameters of 75–95 nm. It can be said that Fe_2O_3 and Fe_3O_4 particles in PCL/PANA provide the formation of more homogeneous and smaller diameter fibers.

Conclusion

Nanofiber films of PCL, PCL/ Fe_2O_3 , PCL/PANA, PCL/PANA₂, PCL/PANA/ Fe_2O_3 , and PCL/PANA/ Fe_3O_4 were successfully prepared by electrospinning. The incorporation of Fe_2O_3 and Fe_3O_4 into nanofibers was confirmed by both SEM and ATR-FTIR. When the thermoelectric properties of nanofibers were examined, it was seen that the power factors of PCL/PANA/ Fe_2O_3 and PCL/PANA/ Fe_3O_4 samples were higher than PCL/PANA samples. It was seen that PANA increased the electrical conductivity, more when the

electrical conductivity of PCL/PANA and PCL/ Fe_2O_3 were compared. There was little difference between the electrical conductivities of PCL/PANA/ Fe_2O_3 and PCL/PANA/ Fe_3O_4 . The S was also higher in nanofibers with Fe_2O_3 and Fe_3O_4 . As a result, the highest S of free-standing and flexible nanofibers was obtained as 49 $\mu\text{V/K}$ from PCL/PANA/ Fe_3O_4 . This result obtained is higher than that of many flexible nanofibers in the literature. The resulting free-standing flexible films can show great potential applications in thermoelectric materials as well as electronic materials and devices.

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Declarations

Conflict of interest The author declares that there is no conflict of interest.

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